3~16 串高精度电池管理系统监测与保护芯片

1 特征

- 设计高度集成，提升系统性能和可靠性
  - 支持 3~16 串电池电压监测功能，满足多样化应用需求
  - 电芯电压测量范围覆盖 1.5V 至 5.0V，适配多种电池类型
  - 可同步测量电压与电流
  - ±200mV 量程电流检测电压范围
  - 支持奇偶均衡，最多支持 8 串均衡功能，内置均衡 MOSFET 导通电阻 25Ω
  - 集成电荷泵，用于高侧充电与放电控制
  - 提供通信接口选项：支持 400kHz I2C 和 2MHz SPI 协议
  - 三路 LDO 独立供电，分别用于 MCU 主电源、接口/无线模组以及 MCU VBAT 的低功耗睡眠电源
  - OTP 存储器支持生产线编程，为设备配置和定制提供灵活性
- 精准的监测能力
  - 全温范围电压监测精度达±7.5mV（单 CELL 刷新周期为 1ms）
  - 全温范围电流检测精度达±50μV（≤5mV 时）
  - 全温范围电流检测精度达±1%（>5mV 时）
  - 电压与电流同步采样，刷新周期 1ms
  - NTC 温度测量精度可达±2°C
- 在恶劣环境下安全可靠运行
  - CELL 1-16 支持高达 85V 的耐压，可支持 CELL 的乱序上电
  - 内置三端保险丝驱动，实现二级保护
  - 配备多种硬件保护机制，包括电压、温度、电流及内部诊断功能
  - 提供多级故障警报和状态指示功能
- 支持多种低功耗运行模式
  - Sleep 模式，功耗为 60μA
  - DeepSleep 模式，功耗为 20μA
  - ShutDown 模式，功耗为 5μA

2 应用场景

- 便携式电源解决方案
- 低压家庭储能应用
- 锂电池驱动的两轮车系统
- 高压电动工具供电
- 用于各类工业电池组的应用场景

3 描述

GD30BM2016 是一款高度集成的电池监控和保护器，专为锂离子电池包设计。该器件能够精确监测 3-16 串电池组的单体电压、电流和温度，同时支持高边充放电功能。它集成了高精度监控系统以及高度可配置的保护子系统，不仅确保监控的精确性，还简化了系统设计。通过提供过充和过放保护，GD30BM2016 有效增强了电池的安全性和性能。

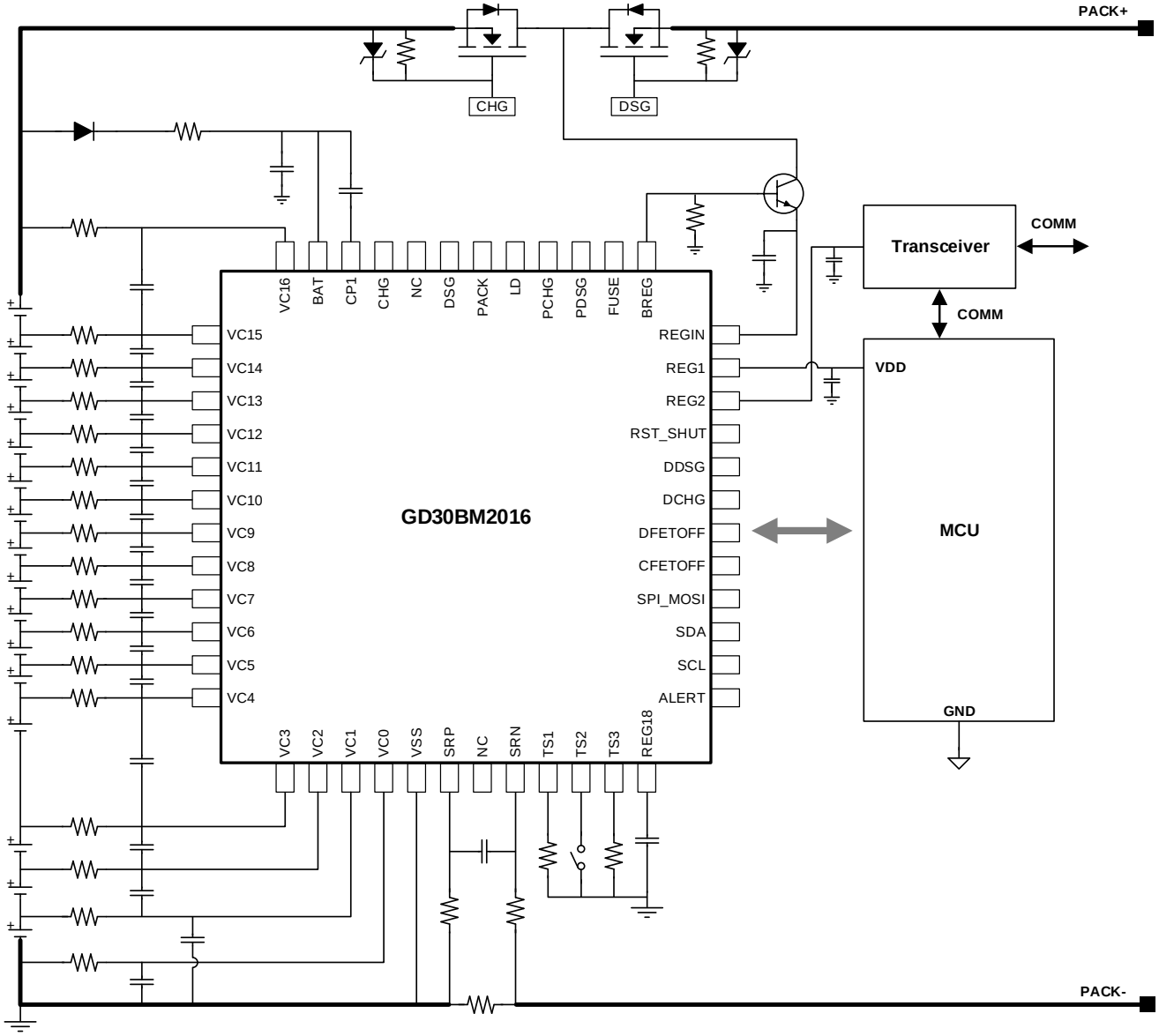
器件信息 [1]

料号	封装	尺寸（标准）
GD30BM2016	LQFP-48	7.00 mm x 7.00 mm

1. 封装详细内容请查看*封装信息*章节。

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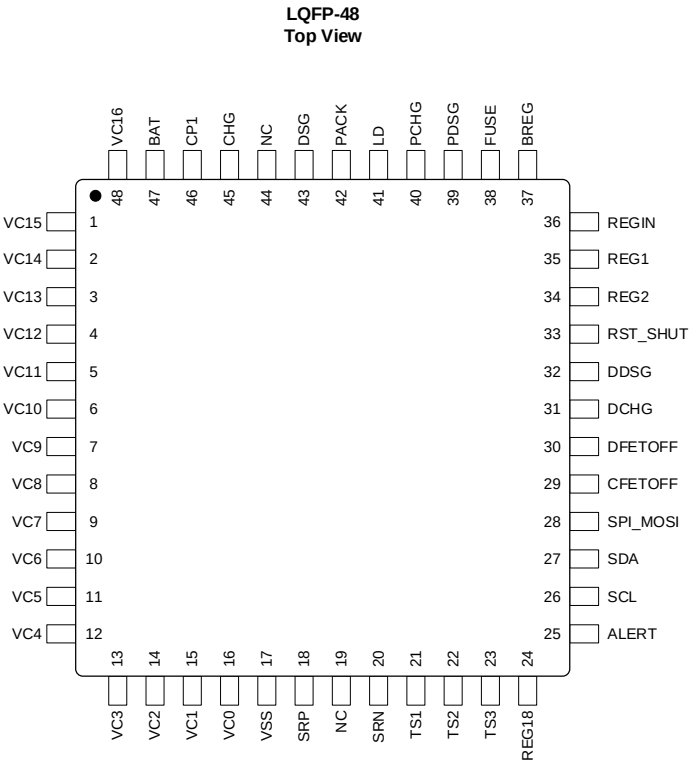
简化电路原理图



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4 引脚配置及功能描述

4.1 引脚分配



4.2 引脚描述

引脚		类型 [1]	功能描述
名称	编号		
VC15	1	I	第十五节电池监测电压输入引脚；第十五节电池平衡电流输入引脚；第十六节电池平衡电流输出引脚
VC14	2	I	第十四节电池监测电压输入引脚；第十四节电池平衡电流输入引脚；第十五节电池平衡电流输出引脚
VC13	3	I	第十三节电池监测电压输入引脚；第十三节电池平衡电流输入引脚；第十四节电池平衡电流输出引脚
VC12	4	I	第十二节电池监测电压输入引脚；第十二节电池平衡电流输入引脚；第十三节电池平衡电流输出引脚
VC11	5	I	第十一节电池监测电压输入引脚；第十一节电池平衡电流输入引脚；第十二节电池平衡电流输出引脚
VC10	6	I	第十节电池监测电压输入引脚；第十节电池平衡电流输入引脚；第十一节电池平衡电流输出引脚
VC9	7	I	第九节电池监测电压输入引脚；第九节电池平衡电流输入引脚；第十节电池平衡电流输出引脚

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引脚		类型 [1]	功能描述
名称	编号		
VC8	8	I	第八节电池监测电压输入引脚；第八节电池平衡电流输入引脚；第九节电池平衡电流输出引脚
VC7	9	I	第七节电池监测电压输入引脚；第七节电池平衡电流输入引脚；第八节电池平衡电流输出引脚
VC6	10	I	第六节电池监测电压输入引脚；第六节电池平衡电流输入引脚；第七节电池平衡电流输出引脚
VC5	11	I	第五节电池监测电压输入引脚；第五节电池平衡电流输入引脚；第六节电池平衡电流输出引脚
VC4	12	I	第四节电池监测电压输入引脚；第四节电池平衡电流输入引脚；第五节电池平衡电流输出引脚
VC3	13	I	第三节电池监测电压输入引脚；第三节电池平衡电流输入引脚；第四节电池平衡电流输出引脚
VC2	14	I	第二节电池监测电压输入引脚；第二节电池平衡电流输入引脚；第三节电池平衡电流输出引脚
VC1	15	I	第一节电池监测电压输入引脚；第一节电池平衡电流输入引脚；第二节电池平衡电流输出引脚
VC0	16	I	第一节电池负极的监测电压输入引脚，第一节电池的平衡电流输出引脚
VSS	17	G	地
SRP	18	I	模拟输入引脚，与内部电流采样和库伦计连接，其中 SRP 为监测电阻上端。充电电流在 SRP 相对于 SRN 产生正电压
NC	19		预留 OTP 烧写使用，接 1uF 电容；不用于 OTP 时，NC 不连接
SRN	20	I	模拟输入引脚，与内部电流采样和库伦计连接，其中 SRN 为监测电阻下端。充电电流在 SRP 相对于 SRN 产生正电压
TS1	21	I/O	热敏电阻输入，或通用 ADC 输入
TS2	22	I/O	用作关机状态唤醒，热敏电阻输入，或通用 ADC 输入
TS3	23	I/O	热敏电阻输入，或通用 ADC 输入
REG18	24	O	内部 1.8V LDO 输出（仅供内部使用）
ALERT	25	I/O	多功能引脚，可作为 ALERT（警报）输出，或热敏电阻输入，或通用 ADC 输入
SCL	26	I/O	多功能引脚，可作为 SCL 或 SPI_SCLK
SDA	27	I/O	多功能引脚，可作为 SDA 或 SPI_MISO
SPI_MOSI	28	I/O	多功能引脚，可作为 SPI_MOSI，或 I2C 地址，或通用数字输出
CFETOFF	29	I/O	多功能引脚，可作为 CFETOFF， SPI_CS，热敏电阻输入，通用 ADC 输入，或通用数字输出
DFETOFF	30	I/O	多功能引脚，可作为 DFETOFF，BOTHOFF，热敏电阻输入，通用 ADC 输入，或通用数字输出
DCHG	31	I/O	多功能引脚，可作为 DCHG，热敏电阻输入，通用 ADC 输入，或通用数字输出
DDSG	32	I/O	多功能引脚，可作为 DDSG，热敏电阻输入，通用 ADC 输入，或通用数字输出
RST_SHUT	33	I	数字输入引脚，用于复位或关机
REG2	34	O	第二个 LDO (REG2) 输出，可通过编程输出 1.8V, 2.5V, 3.0V, 3.3V, 或 5.0V

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引脚		类型 [1]	功能描述
名称	编号		
REG1	35	O	第一个 LDO (REG1) 输出，可通过编程输出 1.8V, 2.5V, 3.0V, 3.3V, 或 5.0V
REGIN	36	I	REG1 和 REG2 LDO 输入引脚
BREG	37	O	外部 PRE-LDO 的晶体管的基极控制信号
FUSE	38	I/O	三端保险丝检测与检测引脚
PDSG	39	O	预放电 P-MOS 控制
PCHG	40	O	预充电 P-MOS 控制
LD	41	I/O	负载检测引脚
PACK	42	I	充电器检测输入引脚
DSG	43	O	主通道 N-MOS 放电驱动输出引脚
NC	44		NC 不连接
CHG	45	O	主通道 N-MOS 充电驱动输出引脚
CP1	46	I/O	电荷泵电容
BAT	47	I	电池组高侧输入引脚
VC16	48	I	第十六节电池监测电压输入引脚，第十六节电池平衡电流输入引脚，以及电池组顶端测量点

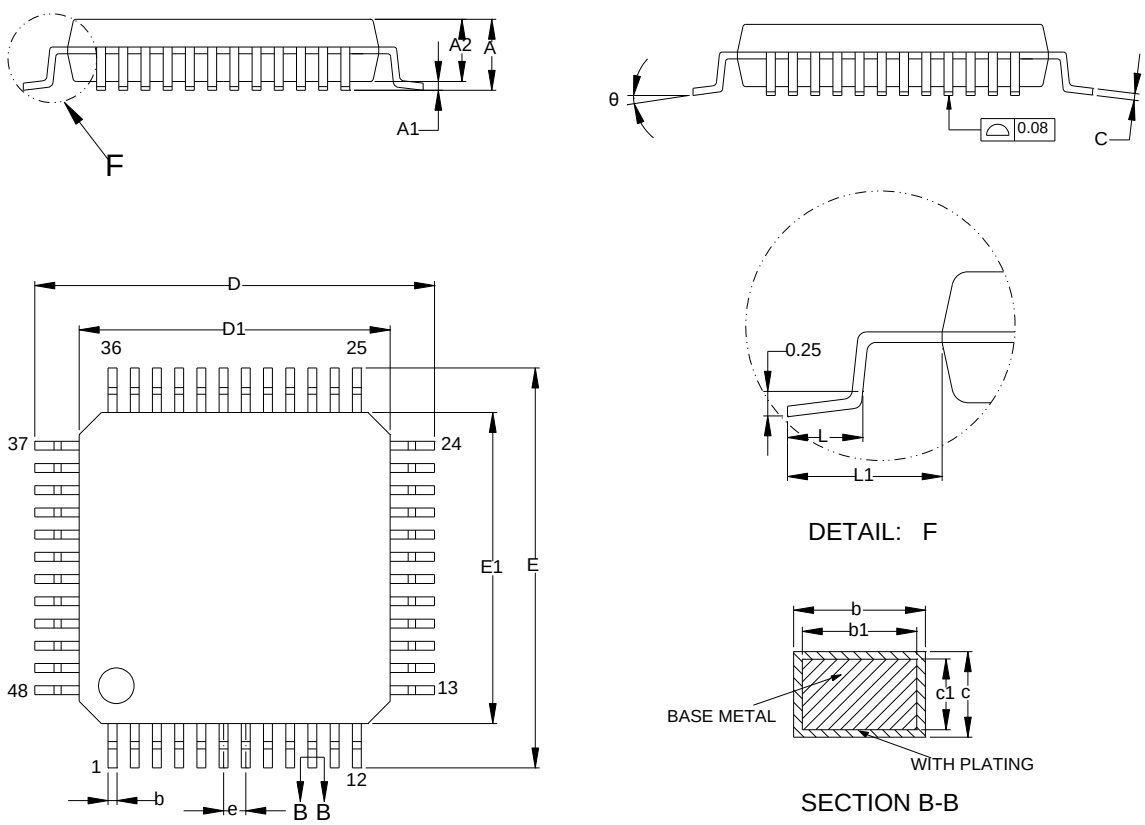
1. I = 输入, O = 输出, P = 电源, G = 地.

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5 封装信息

5.1 外形尺寸

LQFP-48 封装示意图



注:

1. 所有尺寸单位为毫米。
2. 参考表 *1. LQFP-48 尺寸(mm)*。

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表 1. LQFP-48 尺寸(mm)

符号	最小值	典型值	最大值
A			1.60
A1	0.05	0.10	0.15
A2	1.35	1.40	1.45
D	8.90	9.00	9.10
D1	6.95	7.00	7.05
E	8.90	9.00	9.10
E1	6.95	7.00	7.05
C	0.10	0.15	0.20
c	0.09	0.15	0.20
c1	0.124	0.127	0.130
L	0.45	0.60	0.75
L1	1.00REF		
b	0.17	0.22	0.27
b1	0.17	0.20	0.23
e	0.50BSC		
θ	0°	3.5°	7°

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6 产品对比信息

产品型号	通信接口	CRC	REG1 LDO 默认输出电压
GD30BM2016CTTR/TY-I	I2C	N	Disable
GD30BM2016CTTR/TY -I01	SPI	Y	Disable
GD30BM2016CTTR/TY -I02	I2C	Y	Enable set to 3.3V
GD30BM2016CTTR/TY -I03	SPI	Y	Enable set to 5V
GD30BM2016CTTR/TY -I04	SPI	Y	Enable set to 3.3V

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7 订购信息

订购型号	封装类型	ECO Plan	包装类型	最小包装量	温度范围
GD30BM2016CTTR-I	LQFP-48	Green	Tape & Reel	3000	–40°C to +125°C
GD30BM2016CTTR-I01	LQFP-48	Green	Tape & Reel	3000	–40°C to +125°C
GD30BM2016CTTR-I02	LQFP-48	Green	Tape & Reel	3000	–40°C to +125°C
GD30BM2016CTTR-I03	LQFP-48	Green	Tape & Reel	3000	–40°C to +125°C
GD30BM2016CTTR-I04	LQFP-48	Green	Tape & Reel	3000	–40°C to +125°C
GD30BM2016CTTY-I	LQFP-48	Green	Tray	250	–40°C to +125°C
GD30BM2016CTTY-I01	LQFP-48	Green	Tray	250	–40°C to +125°C
GD30BM2016CTTY-I02	LQFP-48	Green	Tray	250	–40°C to +125°C
GD30BM2016CTTY-I03	LQFP-48	Green	Tray	250	–40°C to +125°C
GD30BM2016CTTY-I04	LQFP-48	Green	Tray	250	–40°C to +125°C

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